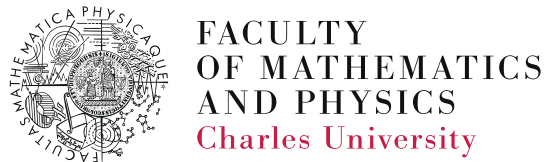


Insights from Graph Interpretation of EASE^R Drive Retrieval Improvements by Uncovering Missed Candidate Items and Correcting Dislike Behavior



Recombee



On Interpretability of Linear Autoencoders

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Introduction

This work addresses the challenge of improving interpretability in linear autoencoders related to EASE^R, which are widely used for candidate retrieval tasks. Despite their strong overall performance, we observed some unexpected behavior in production: (1) **incorporating negative input feedback can lead to inaccurate or counterintuitive recommendations**, and (2) **the models occasionally fail to retrieve a sufficient number of items**.

We introduce a novel graph-based interpretation of EASE^R which reveals **the model's weights are often misinterpreted as indicating either similarity or dissimilarity**. Based on our insights, **we propose a simple modification to EASE^R** that resolves the observed issues and significantly enhances both interpretability and retrieval performance, particularly in datasets with rare item co-occurrence and limited interactions.

Graph-Based Interpretation

Given a user-item interaction matrix X , EASE^R solves the constrained regression problem

$$\min_B \|X - XB\|_F^2 + \lambda \|B\|_F^2 \text{ s.t. } \text{diag}(B) = \vec{0}$$

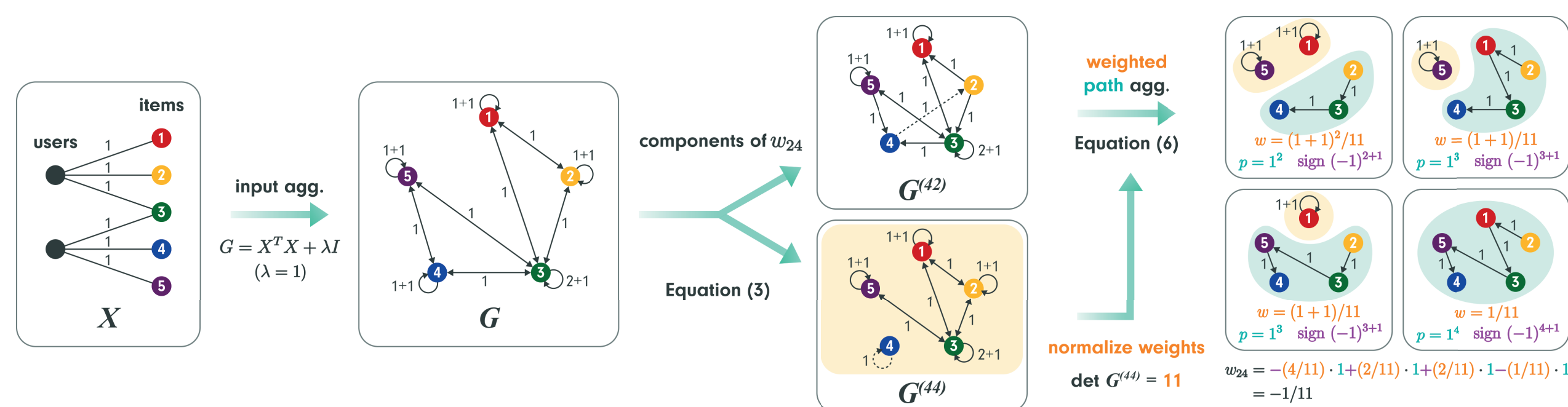
The objective has a closed-form solution:

$$\hat{B} = I - P \text{diagMat}(\vec{1} \odot \text{diag}(P)), \text{ where } P = (X^T X + \lambda I)^{-1}$$

We express $B = W + I$, i.e., as a linear layer with a residual connection. Then, we show that

$$w_{ij} = \sum_c (-1)^{|c|} \frac{\det G_{\mathcal{I} \setminus c}}{\det G_{\mathcal{I} \setminus \{j\}}} \prod_{k \in \mathcal{C} \setminus \{j\}} g_{k,c(k)}$$

The similarity score between item j and item i is calculated as a weighted sum of the product of edge weights, considering every nonzero directed path from i to j in the graph of $G = X^T X + \lambda I$, where each path represents a chain c of observed item co-occurrences.

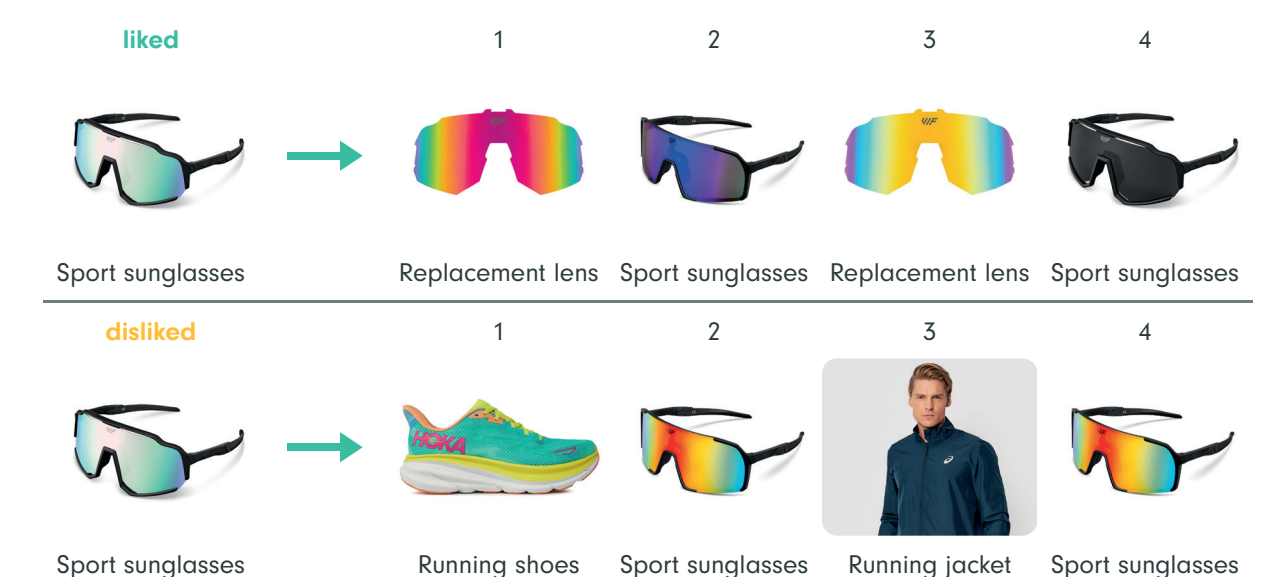


Insights

- **Relevance is global:** Weights w_{ij} aggregate all directed paths from i to j . All item pairs connected in the bipartite graph of X receive a relevance score.
- Each summand (value of a **path/chain**) of w_{ij} consists of 3 components:
 - **bandwidth** (the product of weights along the path)
 - **rarity** (the determinant ratio)
 - **parity** (the sign of the path)

- **Paths pivotal to w_{ij} are short or have high bandwidth.**
- Paths with an even number of edges contribute a negative score.
- **Negative score does not reflect dissimilarity**, merely the parity of dominant paths. Use $\text{abs}(B) = \text{abs}(W) - I$ instead.
- We advise against using negative interactions for training.

Problem



On negative input, EASE^R may promote certain related items instead of hiding them. Being a linear model, this means similar items receive positive or negative scores seemingly at random. Such property is undesirable.

Results

Our approach improves retrieval in sparse domains by:

- **Uncovering previously hidden relevant items.**
- **Correcting issues where items related to dislikes were incorrectly boosted.**

Offline Evaluation

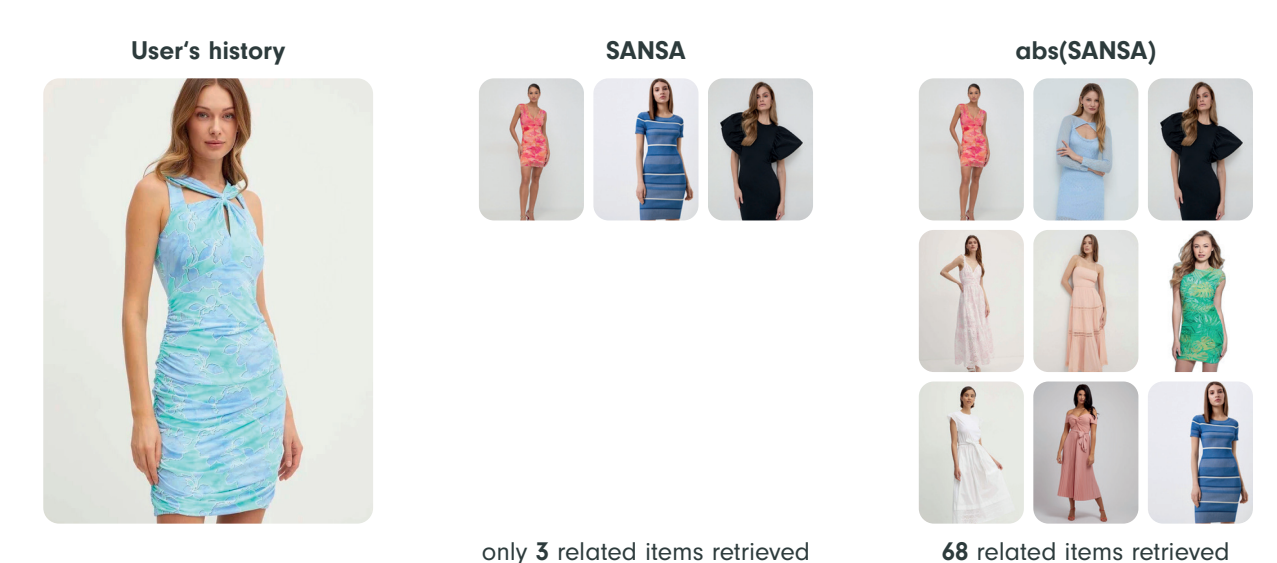
	MovieLens 25M		BeerAdvocate		Yelp2018
nDCG @ 50	P	P+N	P	P+N	P
EASE ^R	0.0774	0.0764	0.1064	0.1052	0.1161
abs(EASE ^R)	0.0825	0.0828	0.1143	0.1123	0.1166
nDCG @ 500	P	P+N	P	P+N	P
EASE ^R	0.1875	0.1853	0.2605	0.2571	0.2500
abs(EASE ^R)	0.2140	0.2194	0.2890	0.2837	0.2637

Using the absolute value of EASE^R weights improves retrieval accuracy by uncovering more items related to the input data. (P and N indicate positive and negative feedback. The Yelp dataset contains only positive feedback, so P+N results are omitted.)

Online Evaluation

A +3% lift in CTR was observed in an e-commerce A/B test.

Case-Based Comparison



New user scenario. Based on a single interaction, SANSA (a sparse variant of EASE^R) could only identify 3 related products among a selection of 4K dresses (out of 2M products). abs(SANSA) retrieved 68 related products.



Inference with negative feedback. While SANSA promoted items similar to the disliked one, we did not observe such behavior with abs(SANSA).



Repository
(dataset,
codes for
reproducibility)



More visual
examples
(case-based
comparisons)

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